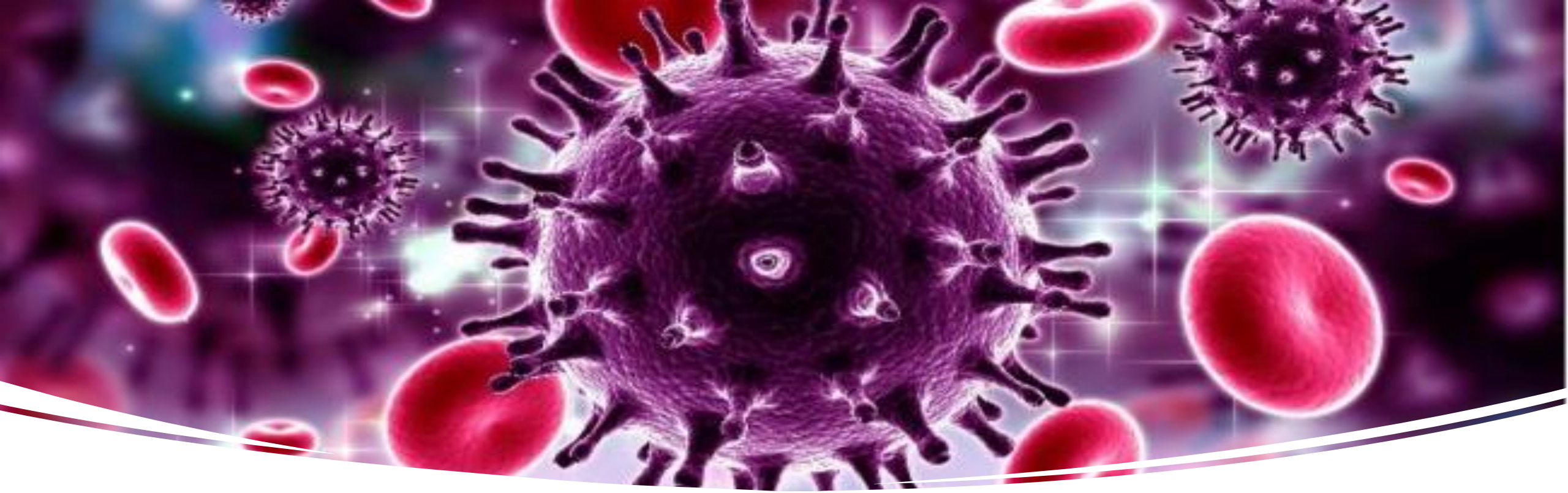
AIDS VIRUS INFECTION PREDICTION

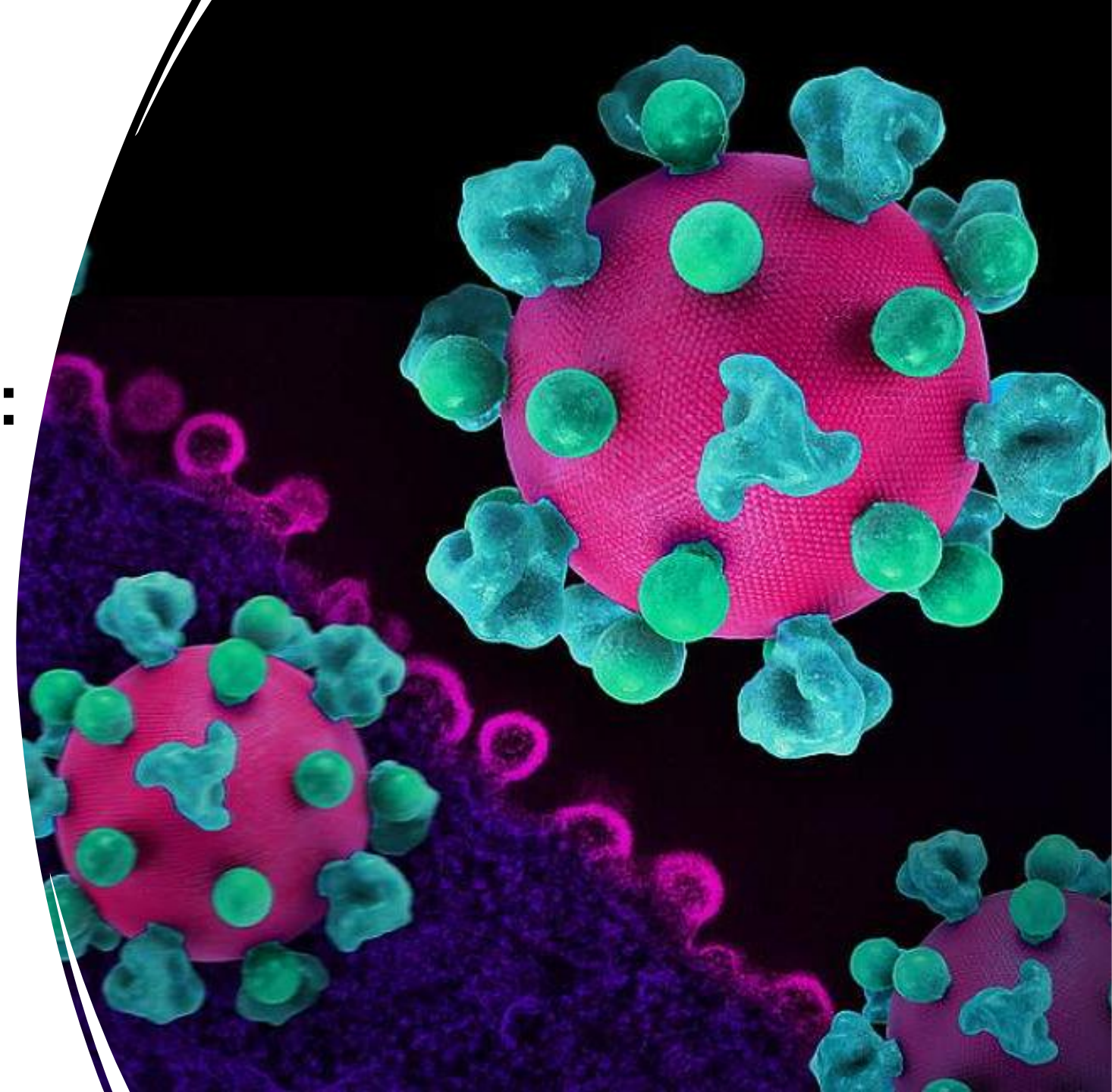
MACHINE LEARNING METHODS AND APPLICATIONS



AIDS VIRUS INFECTION IMPORTANCE

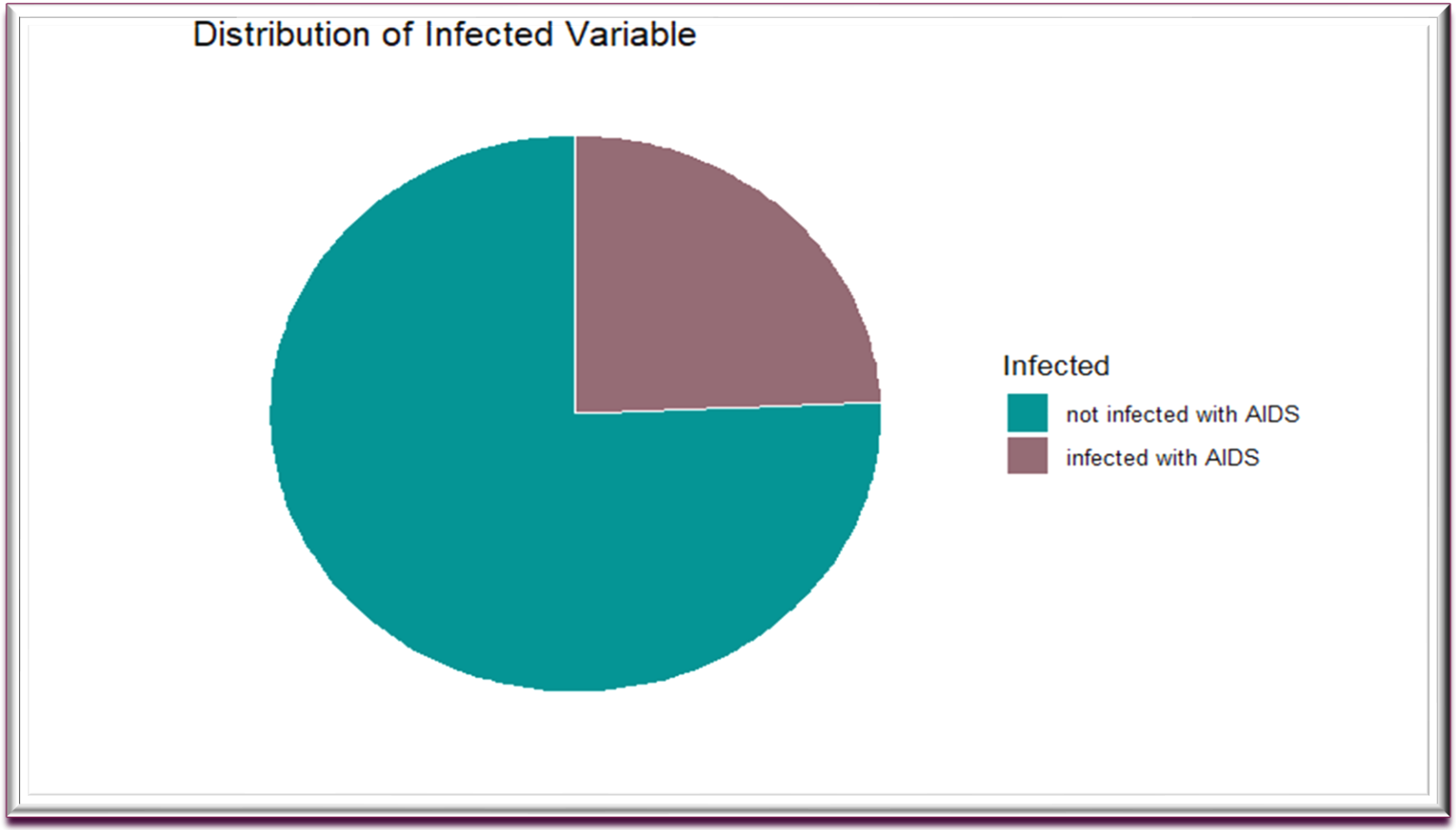
- AIDS (Acquired Immunodeficiency Syndrome) is a disease caused by the HIV (Human Immunodeficiency Virus). It weakens the immune system, making it hard for the body to fight off infections and certain cancers.
- Diagnosing AIDS usually takes a few days to a few weeks, which can be a disadvantage for patients. Also, there is a higher risk of spreading the virus during this time due to patient negligence.
- If the diagnosis process is faster, it benefits patients and saves time and money in healthcare.
- Using machine learning methods can speed up the diagnosis, allowing patients to get treatment sooner and making the healthcare system more efficient.

AIDS VIRUS INFECTION : EXPLORATORY DATA ANALYSIS



TARGET VARIABLE : INFECTED

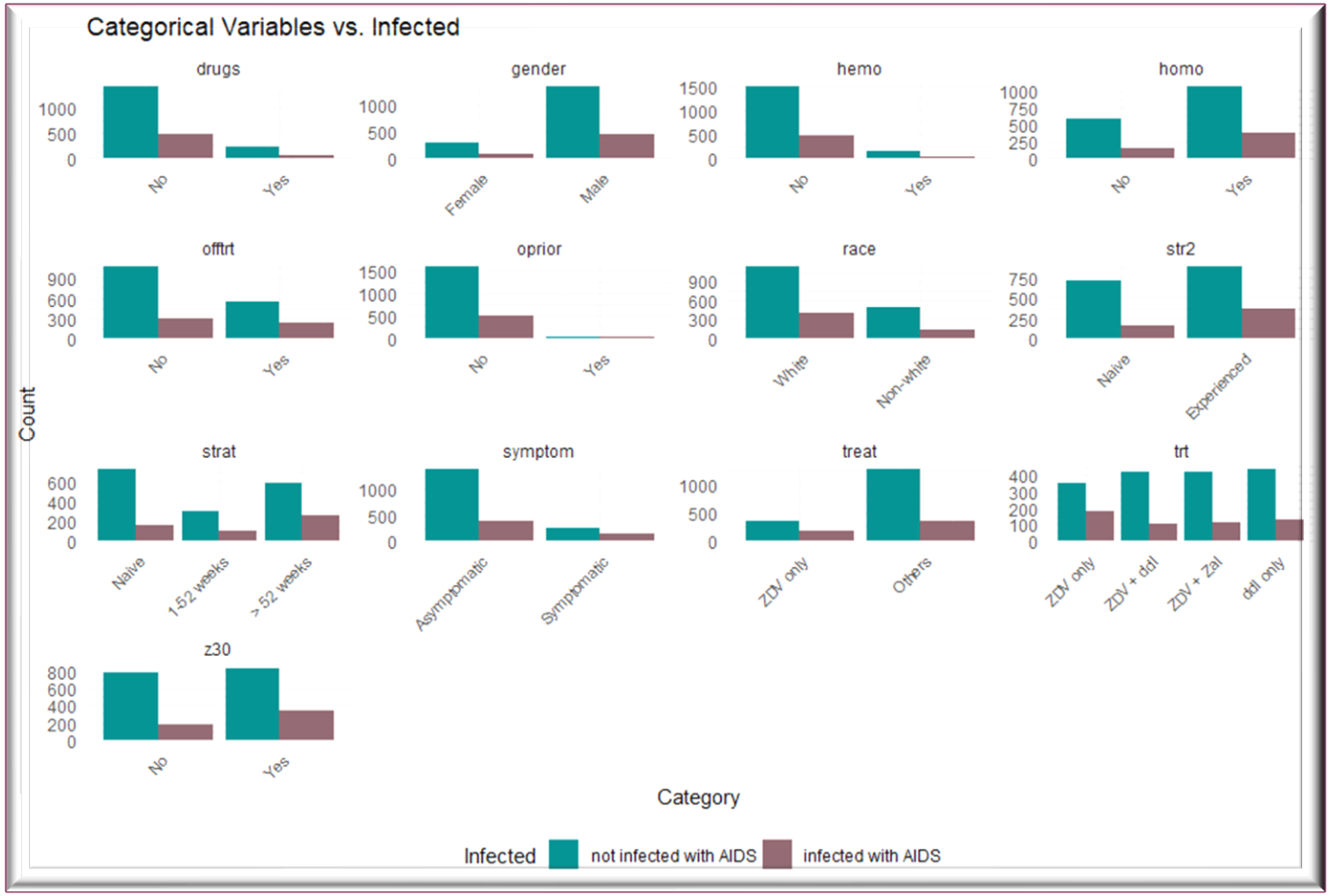
- The dataset contains 2139 observations and 23 variables. The target variable is "infected", and the remaining 22 variables are features.



GRAPH 1: Infected Variable

Definitions and Distributions of Categorical Variables

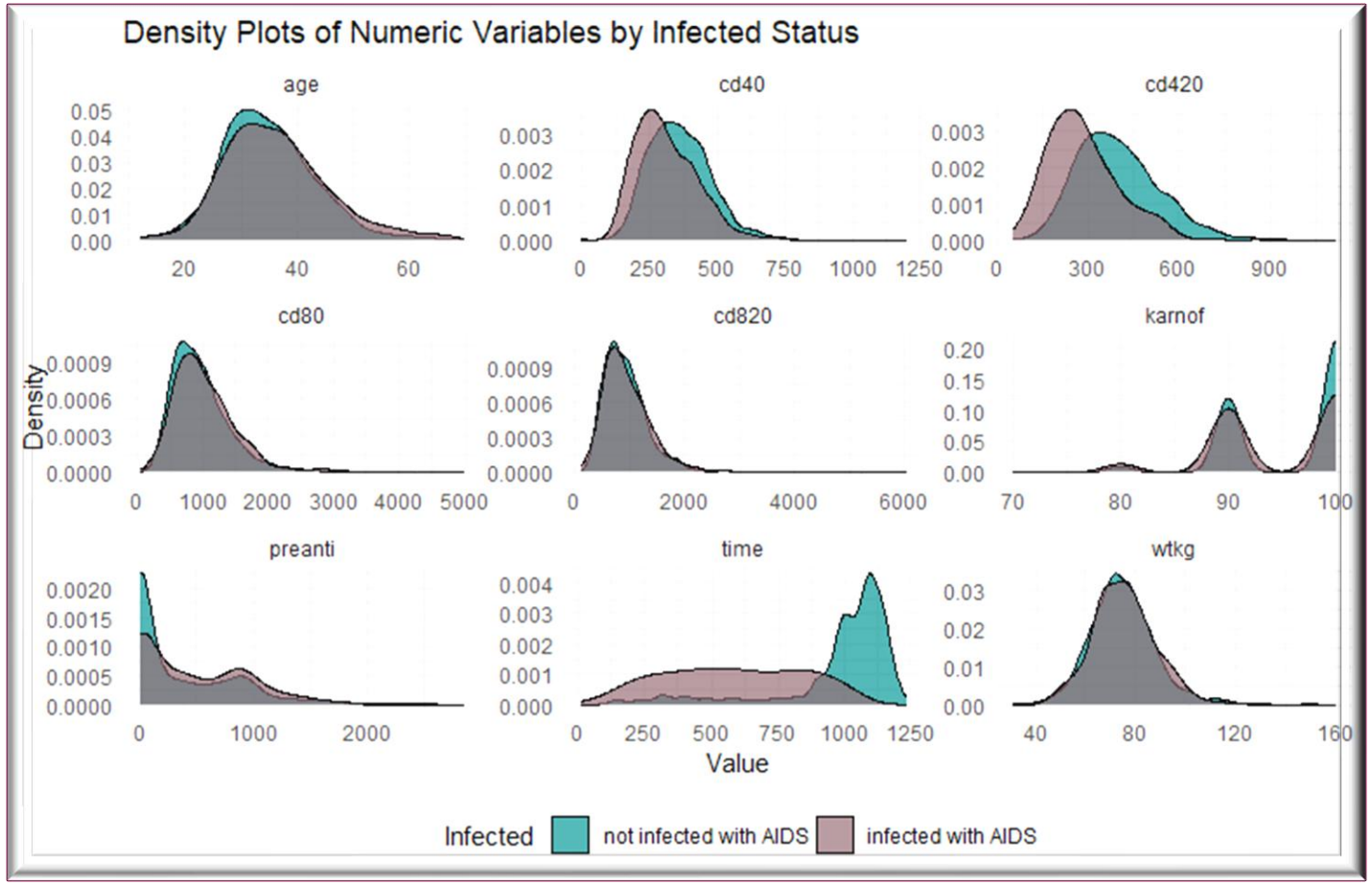
- **drugs:** history of IV drug use (0=no, 1=yes)
- **hemo:** hemophilia (0=no, 1=yes)
- **homo:** homosexual activity (0=no, 1=yes)
- **gender:** gender (0=F, 1=M)
- **offtrt:** indicator of off-trt before 96+/-5 weeks (0=no,1=yes)
- **oprior:** Non-ZDV antiretroviral therapy pre-175 (0=no, 1=yes)
- **race:** race (0=White, 1=non-white)
- **str2:** antiretroviral history (0=naive, 1=experienced)
- **z30:** ZDV in the 30 days prior to 175 (0=no, 1=yes)
- **strat:** antiretroviral history stratification (1='Antiretroviral Naive',2='> 1 but <= 52 weeks of prior antiretroviral therapy',3='> 52 weeks)
- **symptom:** symptomatic indicator (0=asymp, 1=symp)
- **treat:** treatment indicator (0=ZDV only, 1=others)
- **trt:** treatment indicator (0 = ZDV only; 1 = ZDV + ddI, 2 = ZDV + Zal, 3 = ddI only)
- **str2:** antiretroviral history (0=naive, 1=experienced)



GRAPH 2: Categorical Variables

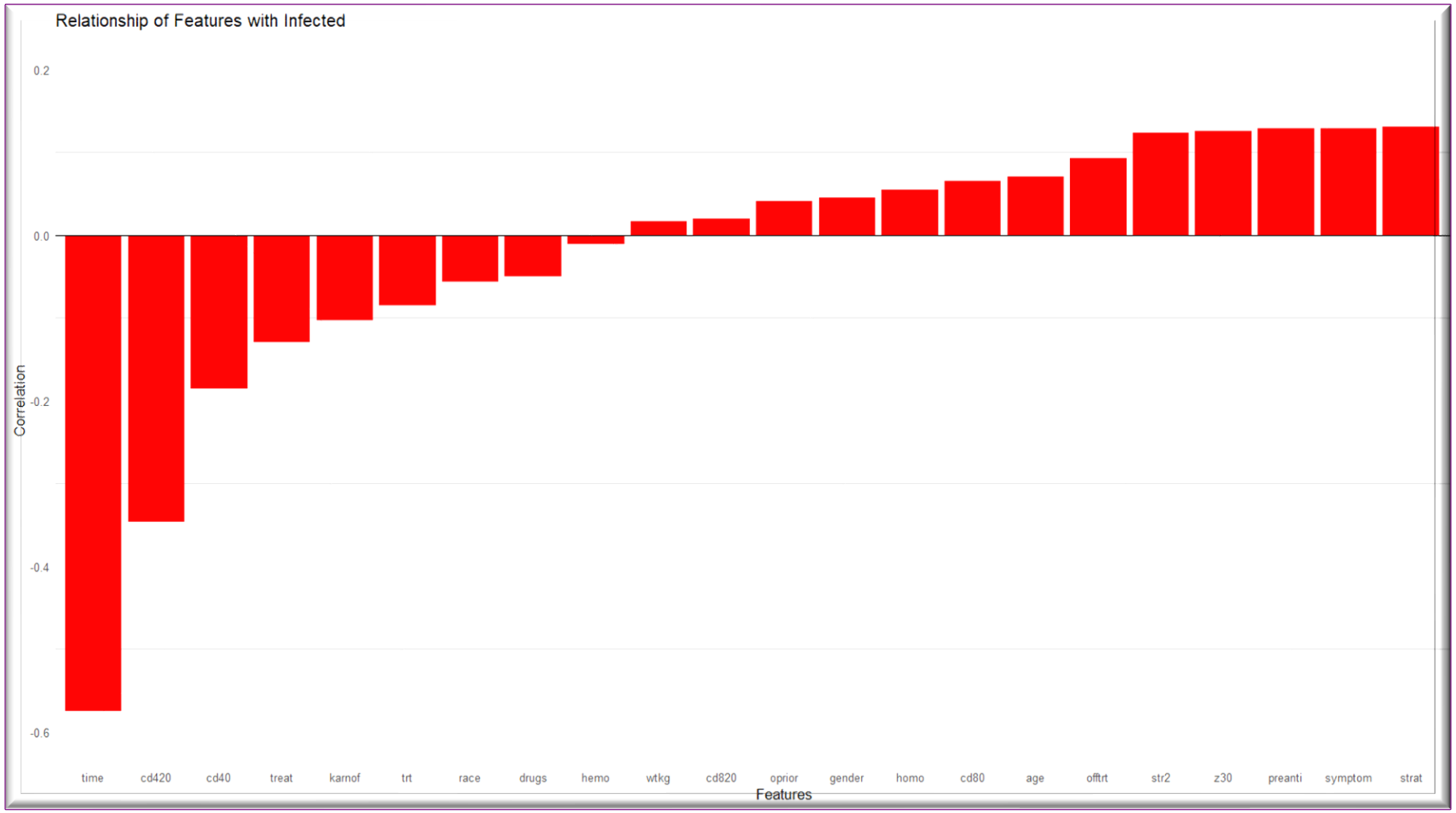
Definitions and Distributions of Numerical Variables

- **time:** time to failure or censoring
- **age:** age (yrs) at baseline
- **wtkg:** weight (kg) at baseline
- **karnof:** Karnofsky score (on a scale of 0-100)
- **preanti:** days pre-175 anti-retroviral therapy
- **cd40:** CD4 at baseline
- **cd420:** CD4 at 20+/-5 weeks
- **cd80:** CD8 at baseline
- **cd820:** CD8 at 20+/-5 weeks



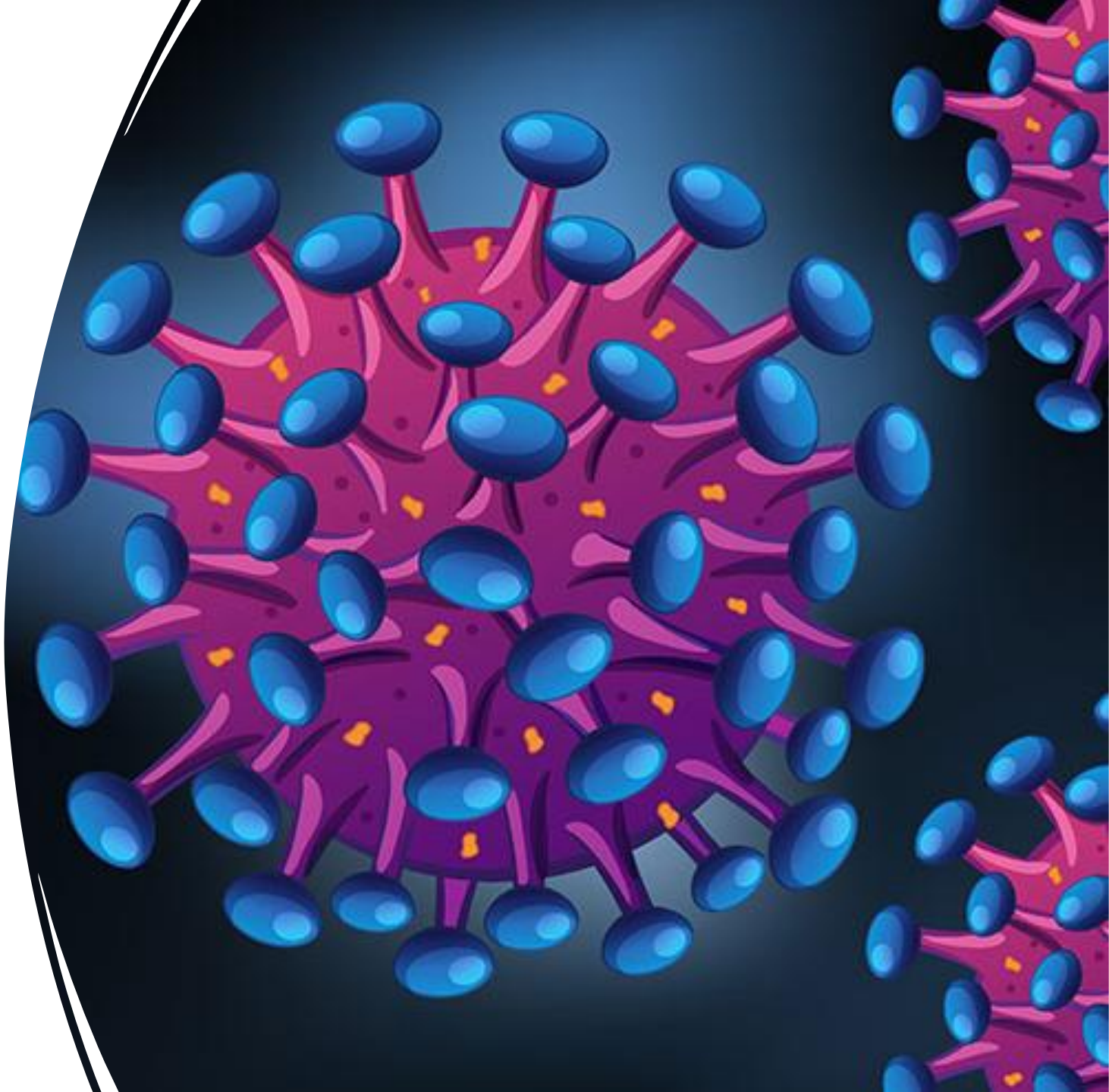
GRAPH 3: Numerical Variables

Correlation of Features with Infected Status

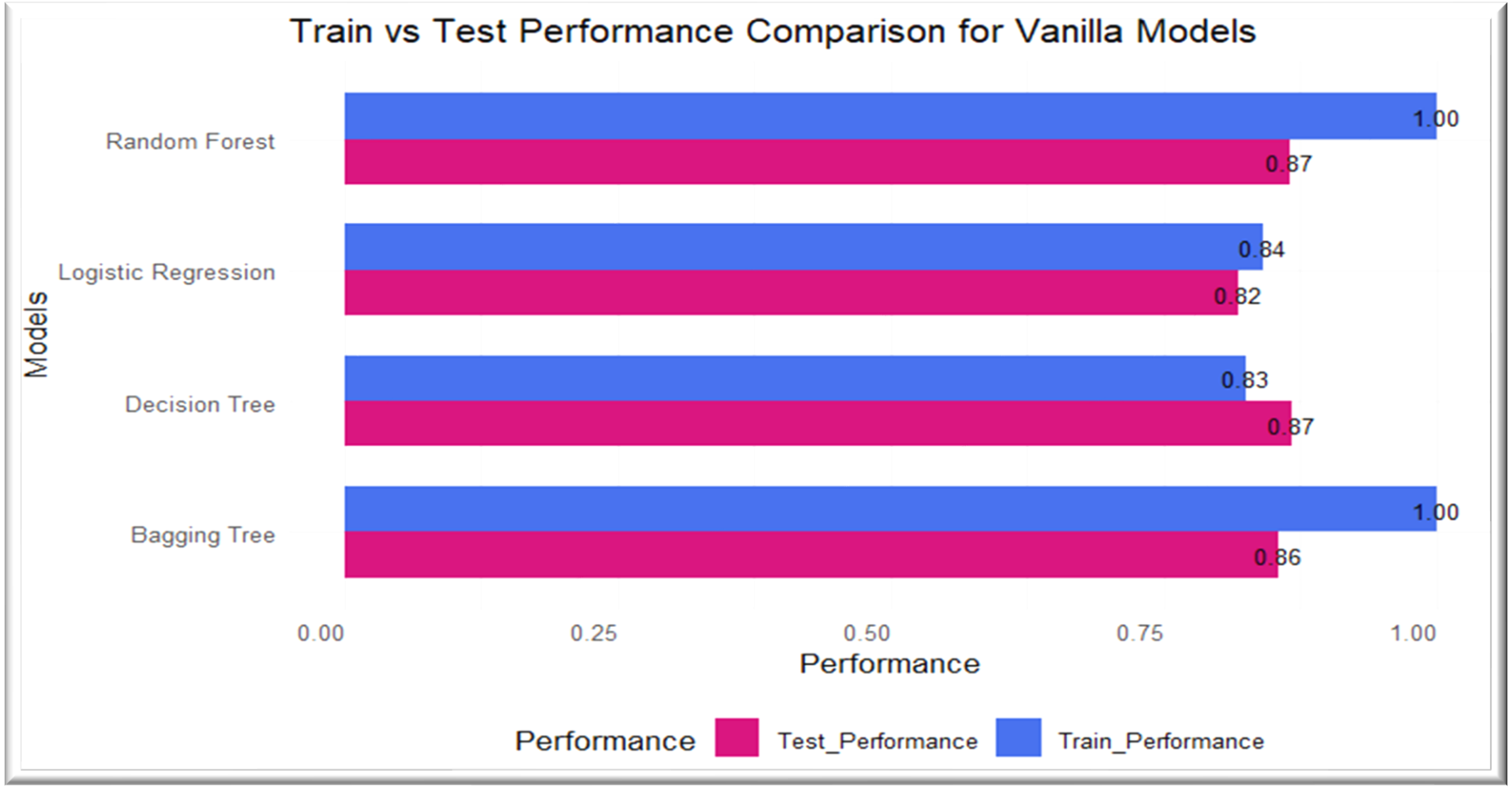


GRAPH 4: Correlation of Features

AIDS VIRUS INFECTION : MACHINE LEARNING MODELS



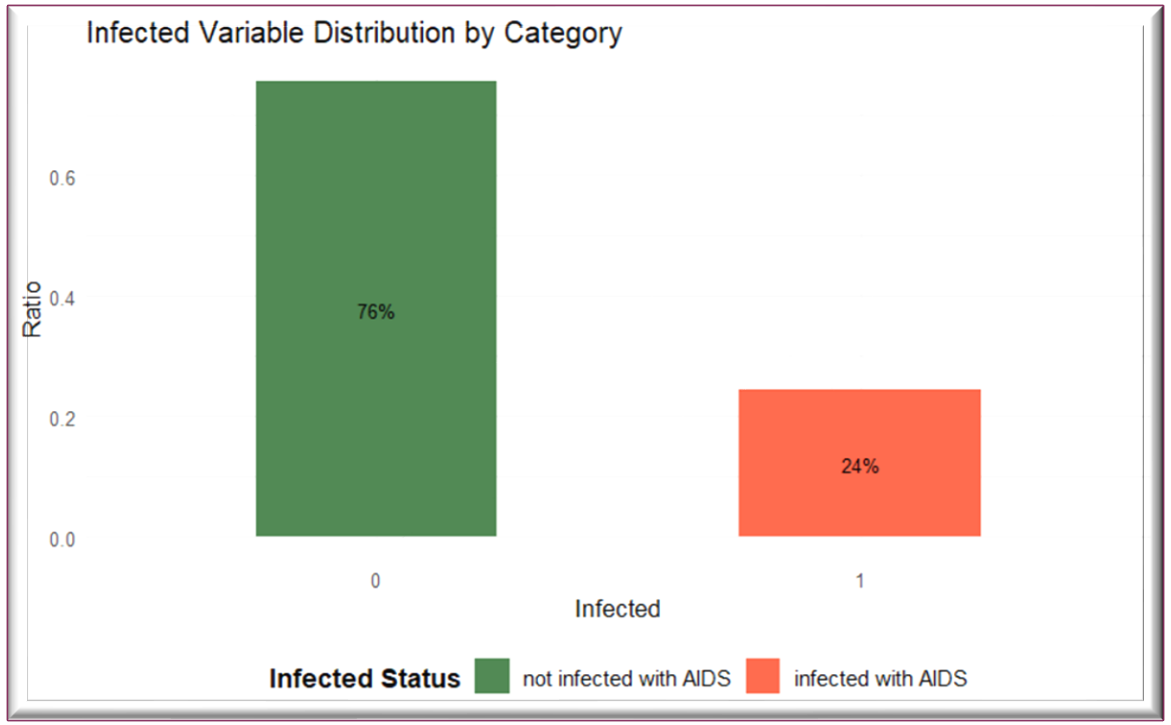
- First, we split the data into training and test sets. Then we compare four different machine learning models, and the conclusions for our models can be seen in the graphic below:



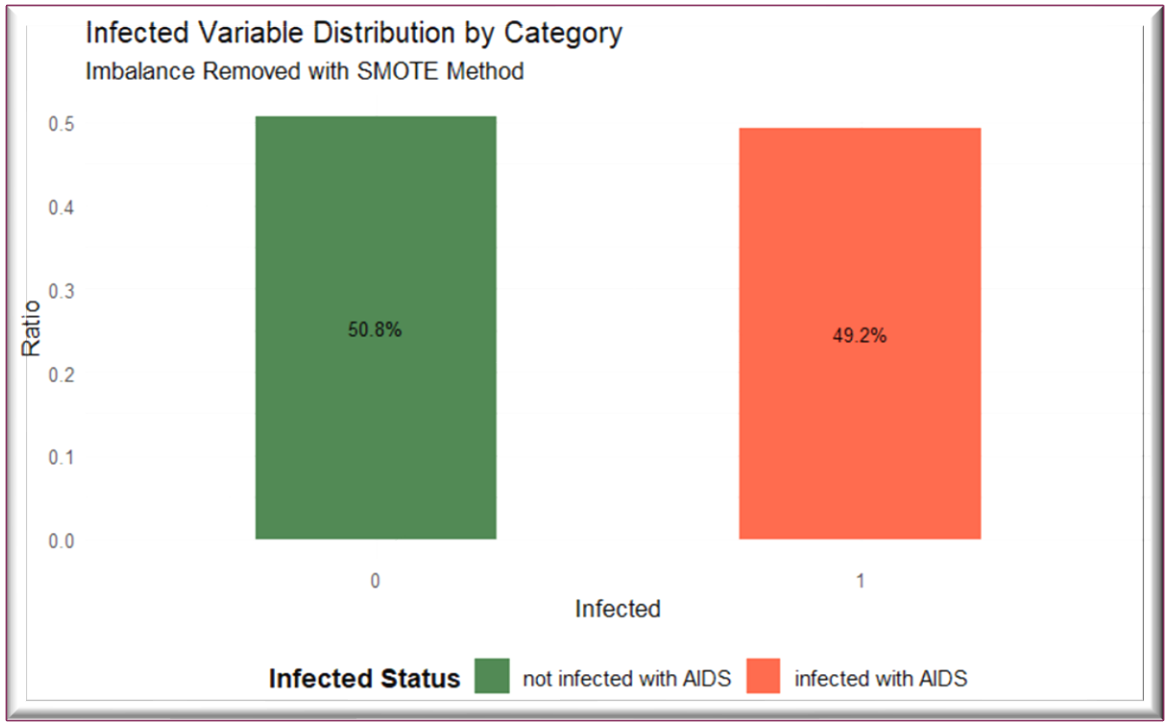
GRAPH 5: Models Performance

Imbalancedness Problem

- Secondly, we check if there is an imbalance problem in our dataset. In graph GRAPH 6, we can clearly see that there is an imbalance, and in graph GRAPH 7, we solve the problem using SMOTE method and we see the corrected version of this issue.



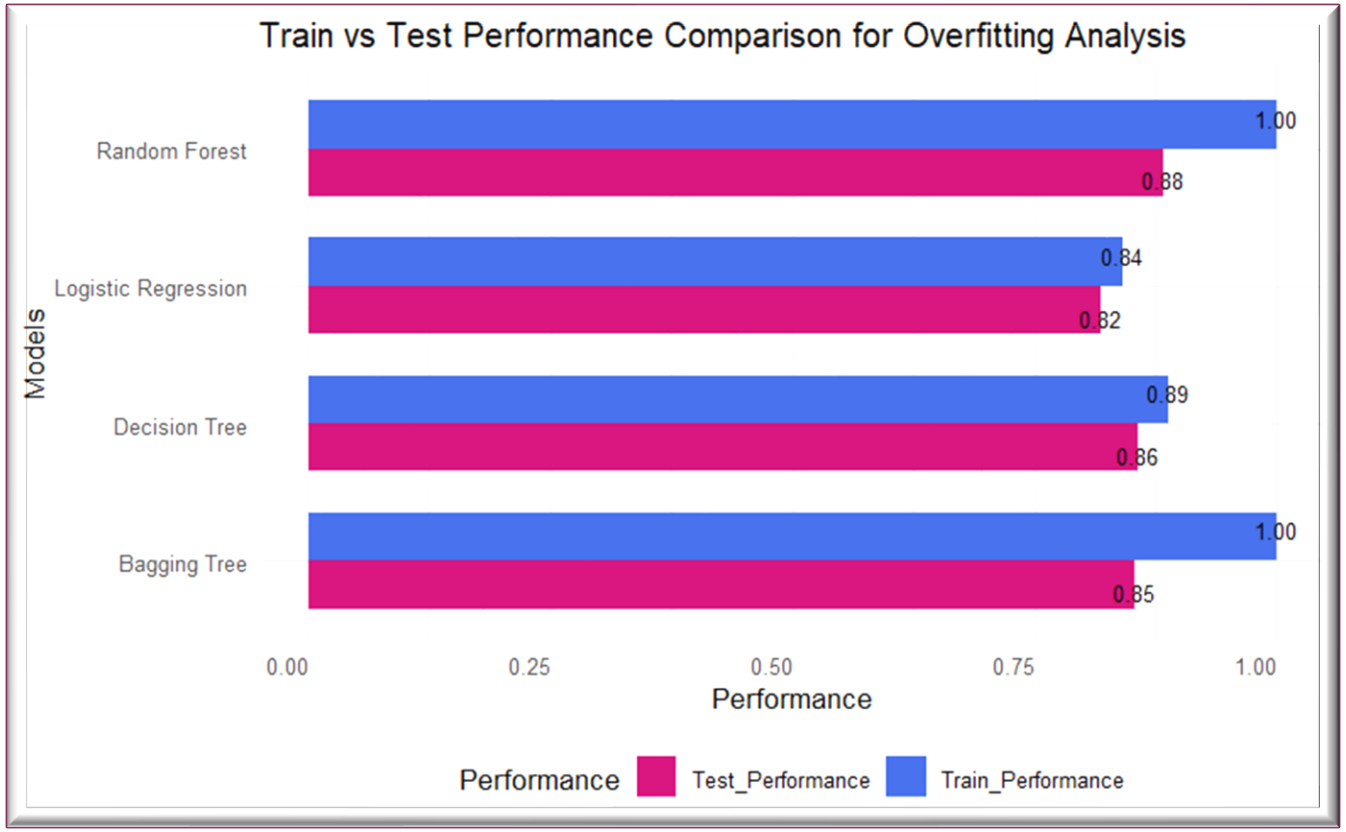
GRAPH 6: Imbalanced of Infected



GRAPH 7: Not Imblance of Infected

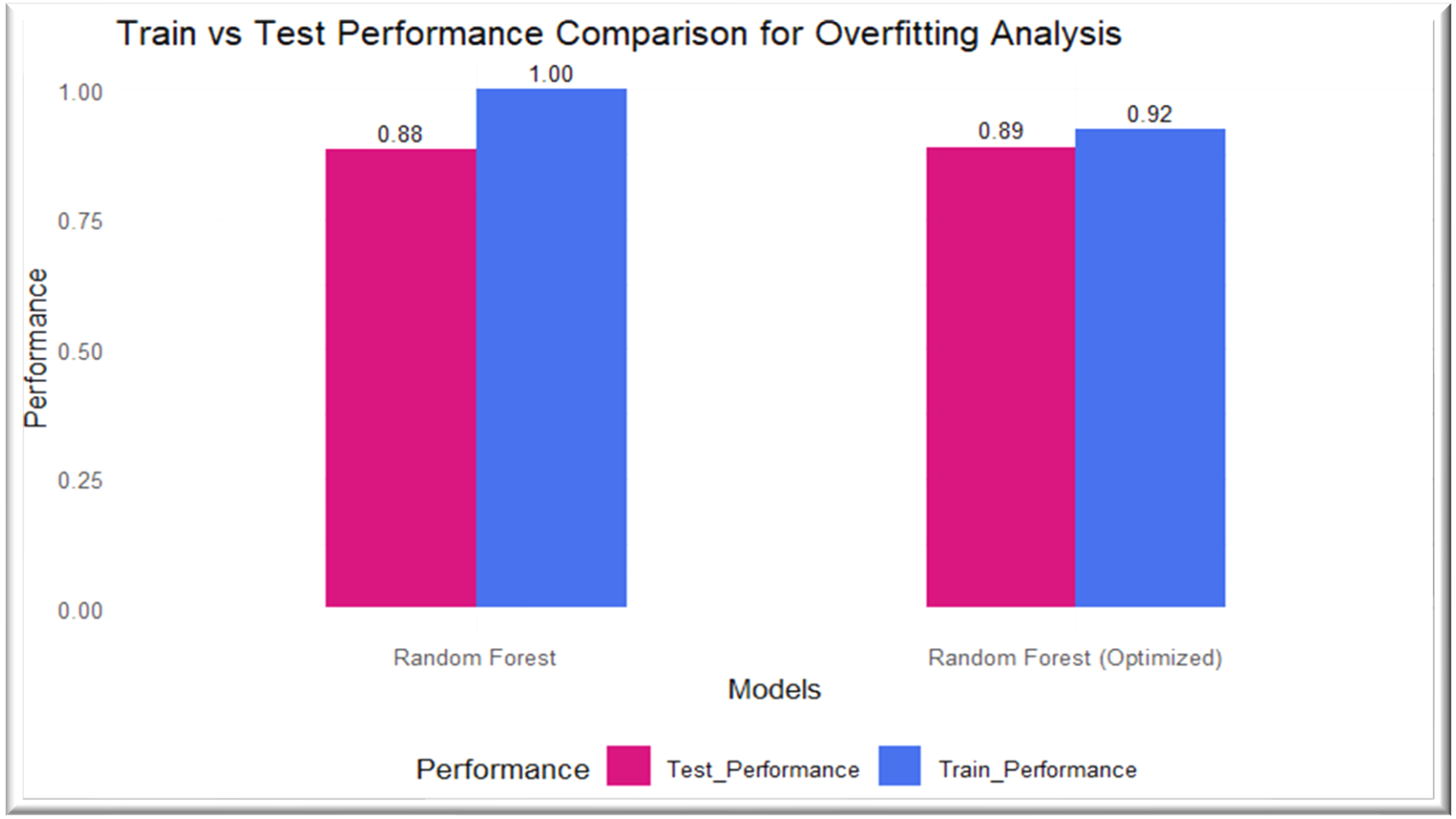
Overfitting Problem

- After solving the imbalance problem, the models were rebuilt. It was observed that the underfitting problem was solved, but as shown in the graph below, the overfitting problem still persists.



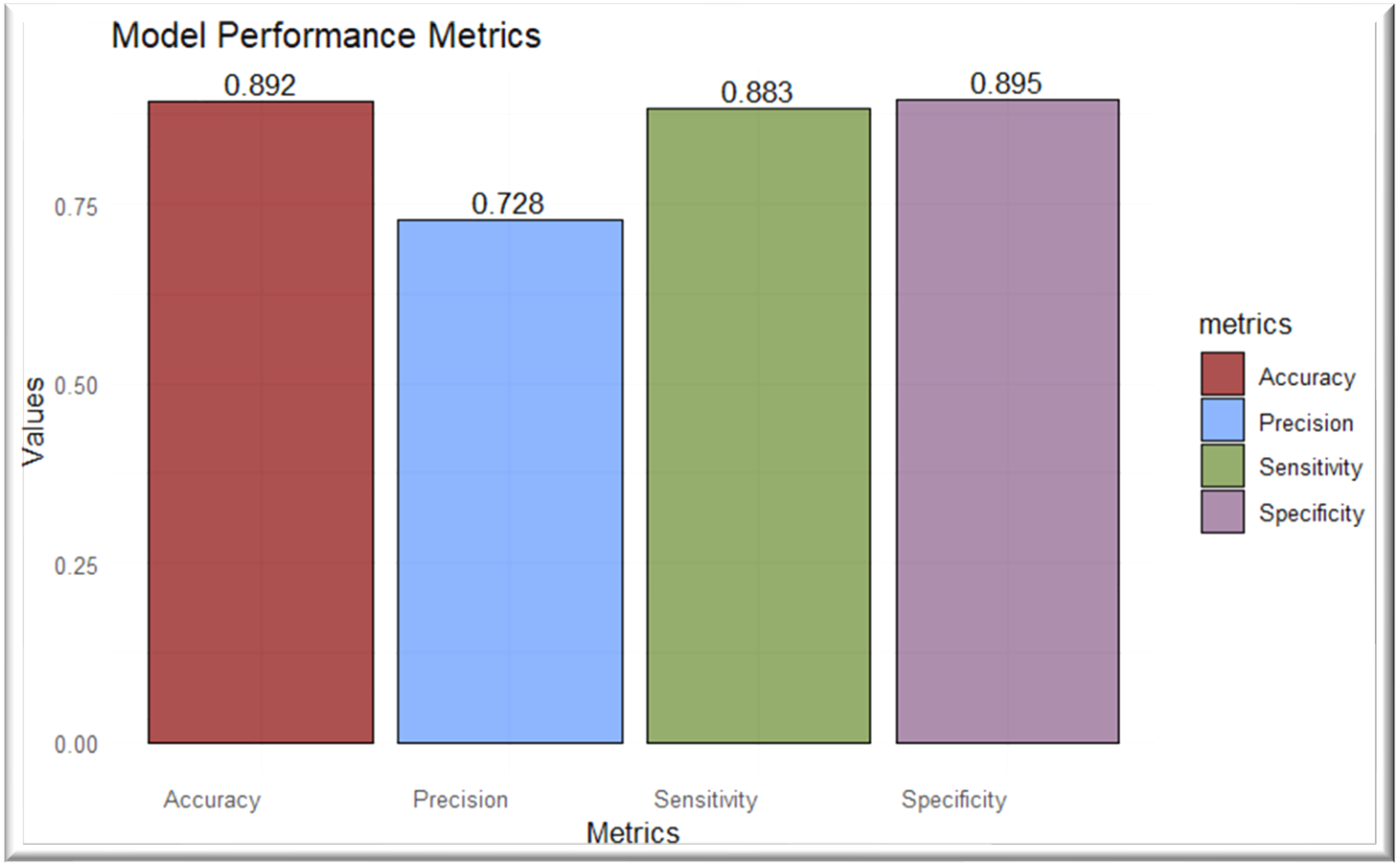
GRAPH 8: Not Imblance of Models

- Out of the four models, the best one is the **Random Forest** model. However, as seen in the GRAPH 9, this model has an overfitting problem. After solving this problem and improving the model's performance, the improved version of the model is shown in the y graph.



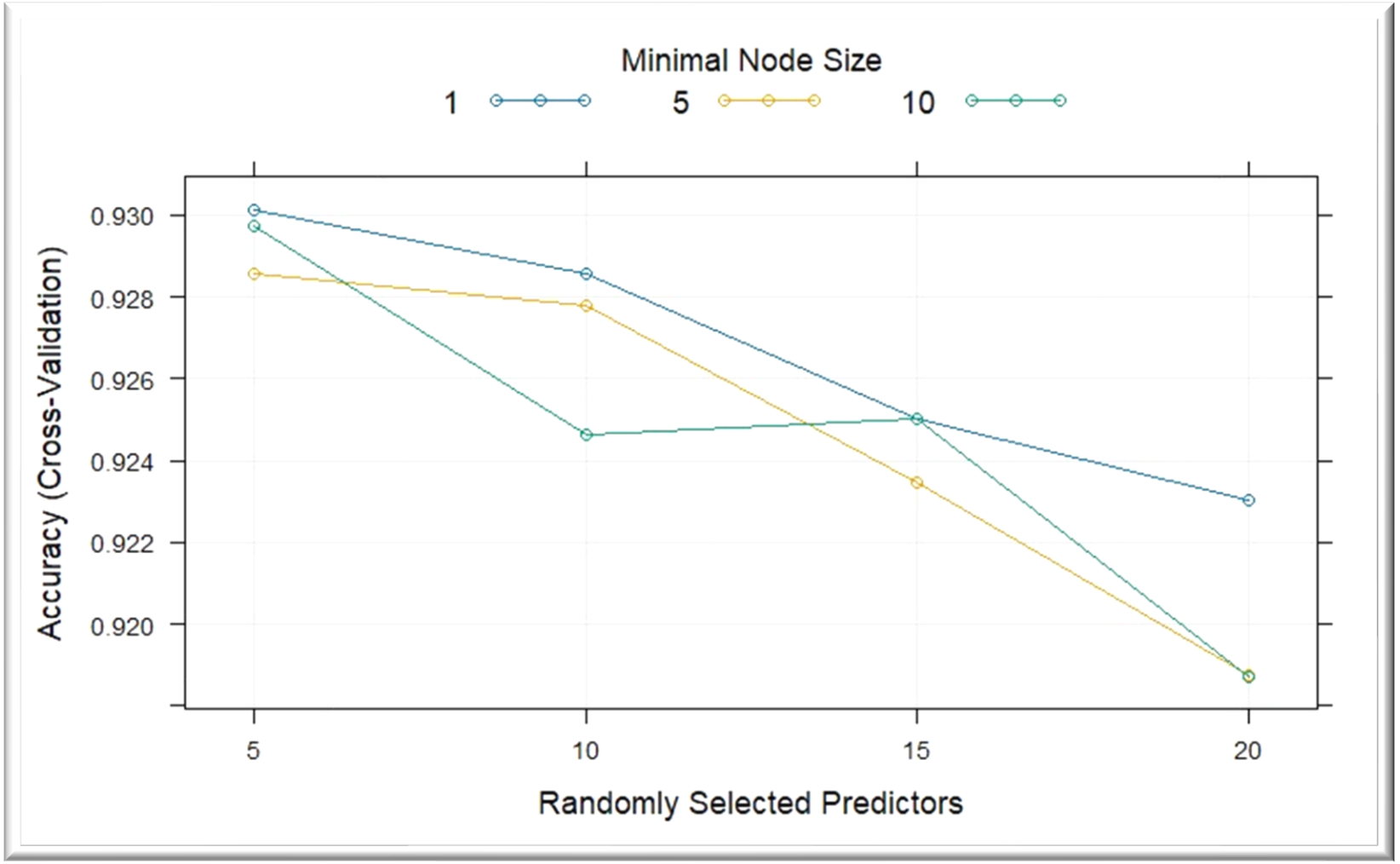
GRAPH 9: Best Model

- Here, the other performance metrics of our Random Forest model are examined, and their values are shown in the x graph below.



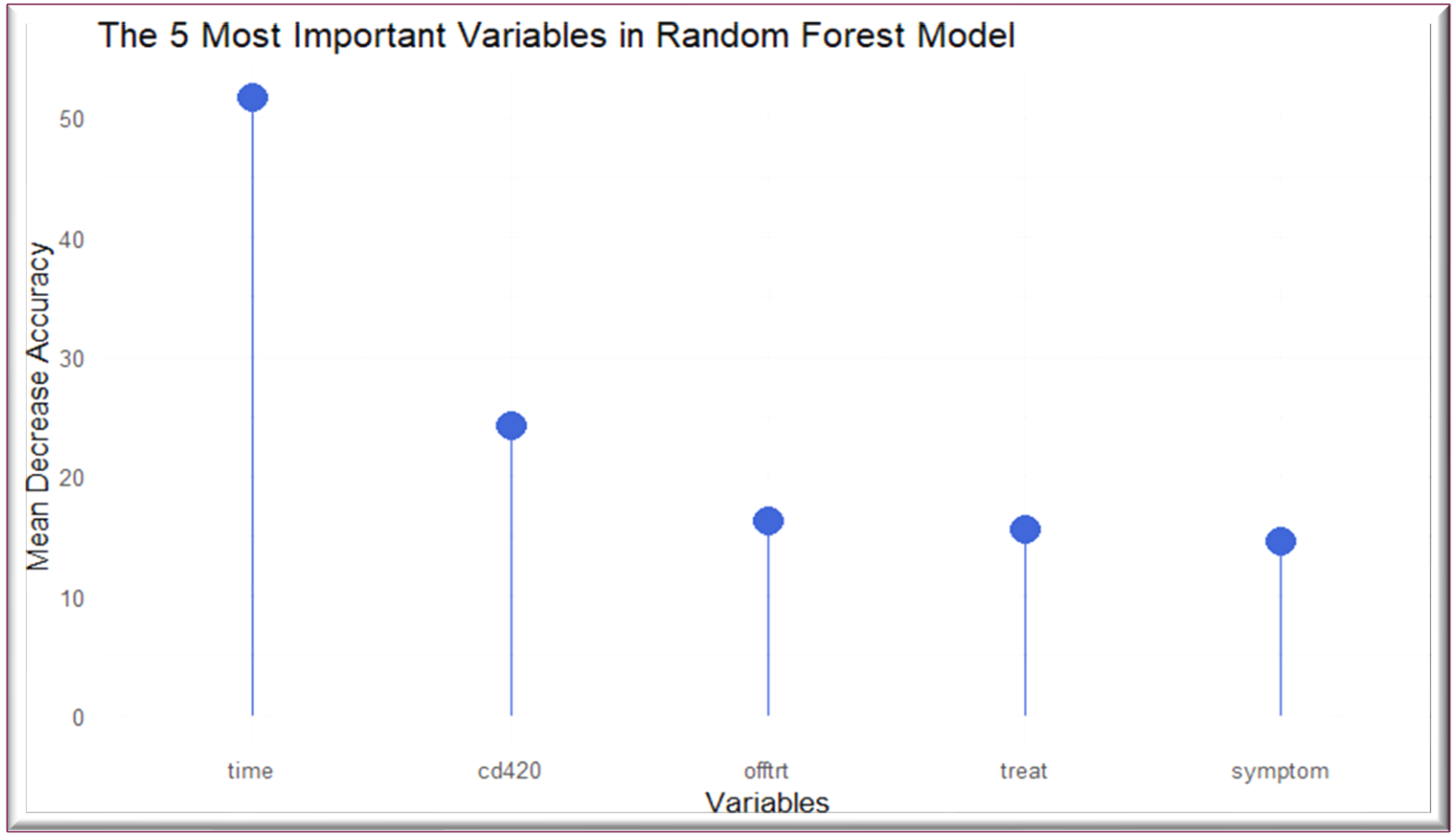
GRAPH 10: Performance Metrics of Random Forest Model

- The graph compares model accuracy (cross-validation) based on the number of randomly selected predictors and minimal node sizes (1, 5, and 10).



GRAPH 11: Cross - Validation

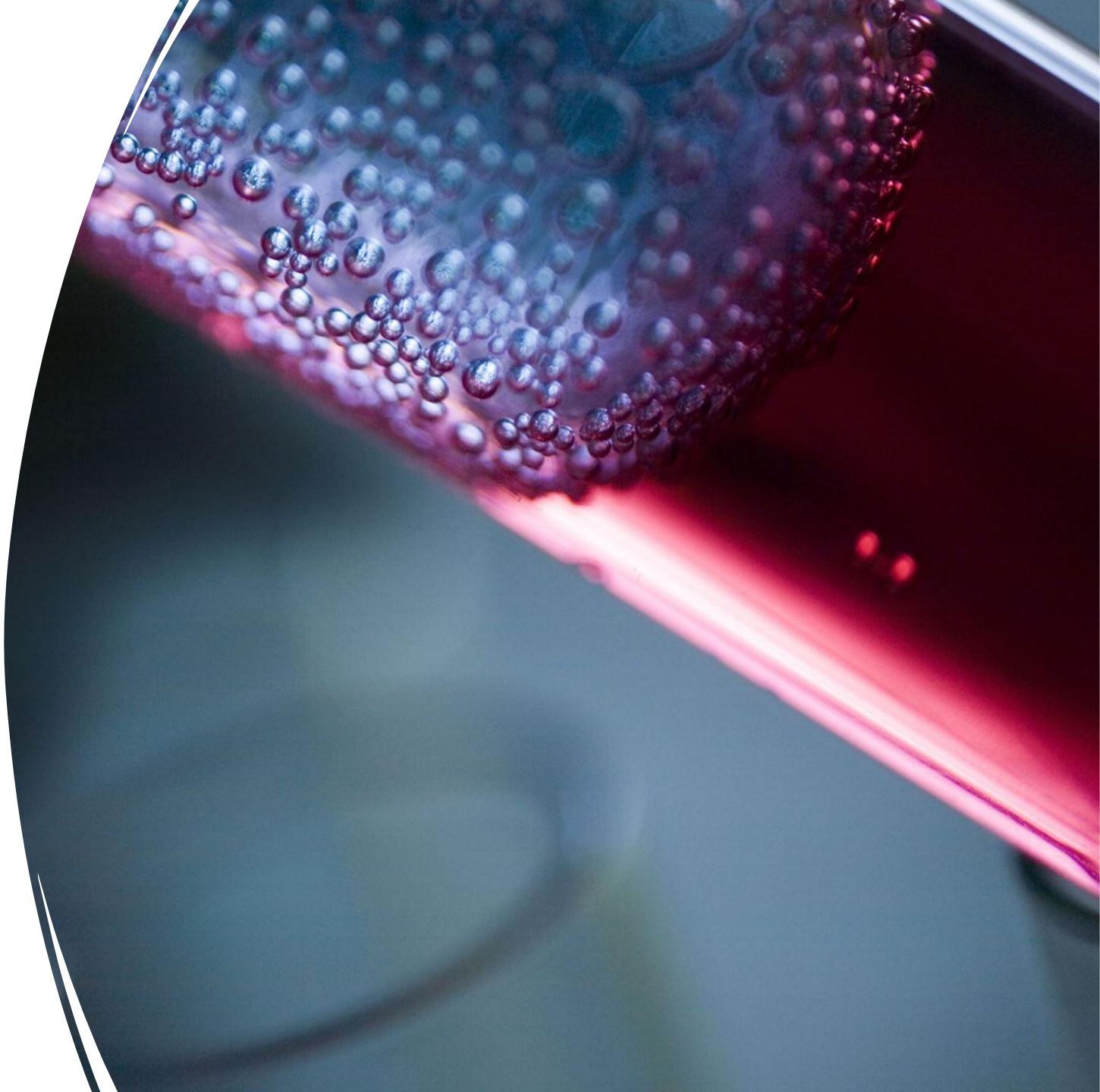
- Lastly, we created a lollipop chart for the top 5 most important variables for our model.



GRAPH 12: Top 5 Important Variable

CONCLUSION

- In conclusion, the model developed to predict individuals infected with AIDS can be further improved to speed up the process in the healthcare sector, allowing patients to start treatment earlier.
- Using machine learning, our model has shown a performance with an 89% success rate and accurate predictions.
- This can enable faster and more accurate diagnoses in healthcare, significantly improving patient care.
- It can optimize the treatment processes for AIDS patients and enhance overall health outcomes.



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2024